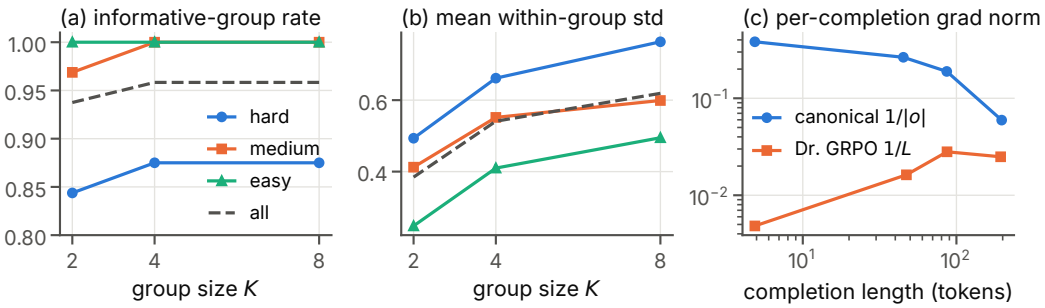
(a) informative-group rate
1.00
0.95
0.90
0.85
0.80
hard
medium
easy
all
2
4
8
group size K
(b) mean within-group std
0.6
0.4
2
4
8
group size K
(c) per-completion grad norm
10^{-1}
10^{-2}
canonical $1/|o|$
Dr. GRPO $1/L$
10^{1}
10^{2}
completion length (tokens)